

L11b: Covered Calls and Protective Collars

CHEME 5660 Cornell University

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Objectives for Today

Today, we analyze two strategies that combine shares with options. What can each trade earn and lose, and how does its value change with the share price?

Today's concepts

- Covered calls: shares plus a short call, two regimes, breakeven, maximum profit and loss.
- Protective collars: shares plus a long put and a short call, three regimes, breakeven, maximum profit and loss.
- Net Delta: the shares plus the weighted Deltas of the option legs, and the shares that make a position delta-neutral.

What You Need to Know

Core concepts

- Covered call profit per share: $\min(S(T), K) - \mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$. The premium lowers the breakeven, and the call caps the profit at the strike.
- Collar profit per share, put at K_1, call at K_2: $S(T) + (K_1 - S(T))^+ - (S(T) - K_2)^+ - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)$. The put sets a floor, and the call a cap.
- Net Delta: $\Delta_{\text{net}} = N_S + \Delta_{\text{opts}}$; holding $N_S = -\Delta_{\text{opts}}$ shares makes a position delta-neutral. A collar's put and call can only lower its net Delta from that of the shares.

For understanding (you won't reproduce these)

- The derivation of the net Delta and the composite Delta table.

Lectures and Examples

Lecture:

CHEME-5660-L11b-Lecture-CoveredCalls-Collars-Fall-2026.ipynb

Examples:

CHEME-5660-L11b-Example-AnalysisCoveredCall-Fall-2026.ipynb

CHEME-5660-L11b-Example-StandardListed-SPY-ProtectiveCollar-Fall-2026.ipynb

Review on your own:

CHEME-5660-L11b-Example-CoveredCallFund-SPY-Fall-2026.ipynb

Company Profile: J.P. Morgan Asset Management

J.P. Morgan Asset Management: the investment management business of JPMorganChase. It runs two large covered call funds, JEPI (S&P 500) and JEPQ (Nasdaq-100):

- Shares plus sold options: each fund owns U.S. large-cap stocks, sells options, and seeks monthly income from the premiums and dividends.
- Equity-linked notes: notes whose value is tied to stocks; as notes, they carry credit or counterparty risk.
- Size: about 45 (JEPI) and 44 (JEPQ) billion USD on September 30, 2026.

Connection: the premiums lower the breakeven, and the sold calls give up gains above the strike.

Covered Calls

A covered call owns shares and sells a call on them, collecting the premium. Investors use it when they expect the share price to stay flat or rise slightly.

Naked calls: selling a call without owning the shares carries unlimited risk if the share price rises. The shares in a covered call cover that obligation.

Covered call ETFs, such as QYLD and FEPI, run this strategy for you.

Video: SMB Capital, Simple & Low Risk Investing Strategy

Covered Call: Profit at Expiration

Buy shares at $S(0)$ and sell one call per m shares, with strike K, premium $\mathcal{P}_c$, and expiration T. The value at expiration minus today's cost grown to expiration is:

$$\begin{aligned}\hat{P}(S(T)) &= S(T) - (S(T)-K)^+ - \mathcal{D}_{T,0}(g_y)\cdot(S(0)-\mathcal{P}_c) \\ &= \underbrace{\big(S(T) - \mathcal{D}_{T,0}(g_y)\cdot S(0)\big)}_{\text{shares}} + \underbrace{\mathcal{D}_{T,0}(g_y)\cdot\mathcal{P}_c - (S(T)-K)^+}_{\text{short call}}\end{aligned}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor and $\hat{P}(S(T))$ is in USD/share. For n contracts, multiply by mn.

Covered Call: Two Regimes

With no dividends and no early exercise, the call's payoff is zero when $S(T) \le K$, so:

$$\hat{P}(S(T)) = \begin{cases} \overbrace{K - \mathcal{D}_{T,0}(g_y) \cdot S(0) + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c}^{\text{realized gain}}, & \text{if } S(T) > K \text{ (exercised)} \\ \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c + \underbrace{\big(S(T) - \mathcal{D}_{T,0}(g_y) \cdot S(0)\big)}_{\text{unrealized gain}}, & \text{if } S(T) \le K \text{ (expires worthless)} \end{cases}$$

- Call exercised: we sell the shares at K, so the whole profit is realized, and the call caps it.
- Call expires worthless: we keep the premium and the shares; the share gain or loss is unrealized.

Covered Call: Breakeven, Maximum Profit, and Maximum Loss

The maximum profit, breakeven $S^{\ddagger}$, and maximum loss are given below. Maximum losses are listed as positive amounts (USD/share):

Maximum profit	$K - \mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$ for $S(T) \geq K$
Breakeven	$S^{\ddagger} = \mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$
Maximum loss	$\mathcal{D}_{T,0}(g_y) \cdot (S(0) - \mathcal{P}_c)$ at $S(T) = 0$

Compared with owning the shares alone, the premium lowers the breakeven, and the call gives up the gains above the strike. If we bought the shares earlier at a cost basis $S^{\dagger}$, the realized gain on the shares at exercise is $K - S^{\dagger}$.

Example: CHEME-5660-L11b-Example-AnalysisCoveredCall-Fall-2026.ipynb **Review on your own:** CHEME-5660-L11b-Example-CoveredCallFund-SPY-Fall-2026.ipynb

Protective Collars

A protective collar adds a long put to a covered call: own the shares, buy a put at K_1 below the share price, and sell a call at $K_2 > K_1$ above it, same expiration T. Per m shares, with premiums $\mathcal{P}_1$ (put) and $\mathcal{P}_2$ (call), the profit at expiration (USD/share) is:

$$\hat{P}(S(T)) = S(T) + \underbrace{(K_1 - S(T))^+}_{\text{long put}} - \underbrace{(S(T) - K_2)^+}_{\text{short call}} - \underbrace{\mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)}_{\text{today's cost, grown}}$$

The put sets a floor under the shares, the call caps them, and the call premium pays for some or all of the put.

Protective Collar: Three Regimes

With no dividends and no early exercise, at most one option pays at expiration:

$$\widehat{P}(S(T)) = \begin{cases} K_1 - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)\,, & \text{if } S(T) < K_1 \text{ (put exercised)} \\ S(T) - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)\,, & \text{if } K_1 \le S(T) \le K_2 \text{ (neither)} \\ K_2 - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)\,, & \text{if } S(T) > K_2 \text{ (call exercised)} \end{cases}$$

- Put exercised: we sell the shares at K_1; the loss is realized and stops growing.
- Neither: we keep the shares; the profit moves with the share price.
- Call exercised: we sell the shares at K_2; the profit is realized and capped.

Protective Collar: Breakeven, Maximum Profit, and Maximum Loss

The maximum profit, breakeven $S^{\ddagger}$, and maximum loss, a positive amount (USD/share), are:

Maximum profit	$K_2 - \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \geq K_2$
Breakeven	$S^{\ddagger} = \mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2)$, if $K_1 \leq S^{\ddagger} \leq K_2$
Maximum loss	$\mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2) - K_1$ for $S(T) \leq K_1$[a]

[a] If $\mathcal{D}_{T,0}(g_y) \cdot (S(0) + \mathcal{P}_1 - \mathcal{P}_2) > K_1$; otherwise none.

When $\mathcal{P}_1 = \mathcal{P}_2$, the call premium pays for the whole put: a zero-cost collar. Compared with the covered call, the put limits the loss, and it costs some of the call premium.

TJUL: A Collar Inside a Fund

The Innovator Equity Defined Protection ETF (TJUL) holds a collar-like package of SPY options from July 1, 2025 to June 30, 2027. Its holdings on September 30, 2026, all expiring June 30, 2027:

- Purchased call, strike 11.09: deep in the money, it moves almost one for one with SPY and stands in for the shares (without dividends).
- Purchased put, strike 616.16: close to the starting SPY level of 617.85, the floor.
- Sold call, strike 718.63: sets the cap.

It seeks a 100% buffer and a 13.61% cap over the whole period, before its 0.79% expense ratio, for holders from first to last day. The cap is on the fund's value, not the call's distance above 617.85.

Example: CHEME-5660-L11b-Example-StandardListed-SPY-ProtectiveCollar-Fall-2026.ipynb **Source:** TJUL fact sheet, September 30, 2026

Delta of Covered Calls and Collars

Previously, we used Delta as an estimate of the probability of exercise. Its other use is hedging: offsetting potential losses with an opposite position in a related counter asset, often the underlying shares.

Hold cash N_C (USD), N_S shares, and the legs of a composite contract $\mathcal{C}$. Leg i has orientation θ_i (1 long, -1 short), n_i contracts, multiplier m, and premium $\mathcal{P}_i(S,t)$ (USD/share). The wealth at time t is:

$$W_t = N_C + N_S\, S + m \sum_{i \in \mathcal{C}} \theta_i n_i \mathcal{P}_i(S,t)$$

The Net Delta of a Portfolio

The net Delta is the sensitivity of the wealth to the share price:

$$\Delta_{\text{net}}(S,t) = \frac{\partial W_t}{\partial S} = \underbrace{\frac{\partial N_C}{\partial S}}_{=0} + \underbrace{\frac{\partial (N_S S)}{\partial S}}_{=N_S} + m \sum_{i \in \mathcal{C}} \theta_i n_i \underbrace{\frac{\partial \mathcal{P}_i}{\partial S}}_{=\Delta_i(S,t)}$$

$$= N_S + \underbrace{m \sum_{i \in \mathcal{C}} \theta_i n_i \Delta_i(S,t)}_{= \Delta_{\text{opts}}(S,t)}$$

The options contribute $\Delta_{\text{opts}} = m\,\Delta_{\mathcal{C}}$, where $\Delta_{\mathcal{C}}(S,t) = \sum_{i \in \mathcal{C}} \theta_i n_i \Delta_i(S,t)$ is the composite Delta.

Delta-Neutral Positions

A position is delta-neutral if $\Delta_{\text{net}}(S,t) = 0$, which we get by holding $N_S = -\Delta_{\text{opts}}(S,t)$ shares. For example, short 10 calls at $\Delta = 0.6$ with $m = 100$ give:

$$\Delta_{\text{opts}} = 100 \times (-1) \times 10 \times 0.6 = -600 \quad \Longrightarrow \quad N_S = 600 \text{ shares}$$

If leg i is exercised or assigned at time $\tau \le T$, set $\Delta_i = 0$ for $t \ge \tau$ and update the shares:

$$N_S \leftarrow N_S + \theta_i m n_i \ \text{(call)} \qquad N_S \leftarrow N_S - \theta_i m n_i \ \text{(put)}$$

Covered Call and Collar Delta

A covered call holds $N_S = mn$ shares and is short n calls with Delta Δ_c; a collar adds n long puts with Delta Δ_p (put at K_1, call at K_2). The cash does not contribute:

$$\Delta_{\text{net}}(S,t) = mn\,(1 - \Delta_c(S,t)) \qquad \text{covered call}$$

$$\Delta_{\text{net}}(S,t) = mn\,\big(1 + \Delta_p(S,t) - \Delta_c(S,t)\big) \qquad \text{collar}$$

Since $0 \le \Delta_c \le 1$ and $-1 \le \Delta_p \le 0$, the put and the call can only lower the collar's net Delta from mn. Delta-neutral cases:

- Covered call: $\Delta_c \approx 1$, the call deep in the money ($S(t) \gg K$).
- Collar: beyond either strike, $S(t) \ll K_1$ ($\Delta_p \approx -1$) or $S(t) \gg K_2$ ($\Delta_c \approx 1$).

Delta of the Common Composite Contracts

Choose the strikes by Delta, a common rule of thumb: short legs at ± 0.25, protecting long legs at ± 0.10, the straddle at the money (about ± 0.5). At entry, for one contract ($m = 100$):

Composite contract	Composite Delta, $\Delta_{\mathcal{C}}$	Shares to hedge, N_S
Short put	$-(-0.25) = 0.25$	-25
Short call	-0.25	$+25$
Put credit spread	$-0.10 - (-0.25) = 0.15$	-15
Call credit spread	$-0.25 + 0.10 = -0.15$	$+15$
Short straddle	$-(-0.5) - 0.5 \approx 0$	≈ 0
Short strangle	$-(-0.25) - 0.25 = 0$	0
Protected strangle	$-(-0.25) - 0.25 + 0.10 = 0.10$	-10
Iron condor	$-0.10 - (-0.25) - 0.25 + 0.10 = 0$	0

A Delta of zero is not zero risk: Gamma moves the Delta as the share price changes, so a delta-neutral position must update N_S.

Summary

We analyzed covered calls and protective collars, two strategies that combine shares with options, and computed the net Delta of shares plus options.

Key takeaways

- Covered calls: the call caps the profit at the strike, and the premium lowers the breakeven.
- Protective collars: the put sets a floor and the call a cap; in between, the position moves with the shares.
- Net Delta: the shares plus the weighted Deltas of the option legs; a collar's put and call can only lower it from that of the shares.

These strategies trade some of the upside of the shares for income or for protection.